# Printing for Web Developers

Tanu Hoque

Noah Sager

*ESRI DEVELOPER & TECHNOLOGY SUMMIT*

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# Printing for Web Developers

YOU ARE A PARTICIPANT

Friday, Mar 13 | 8:30 AM - 9:30 AM PDT

Pasadena | Renaissance Hotel

This technical session focuses on printing as a collaborative technology for web developers. Speakers cover the entire printing workflow, starting with creating and sharing custom print layout templates in ArcGIS Pro, ArcGIS Online, and ArcGIS Enterprise. They then move on to printing in a variety of ArcGIS apps, including Map Viewer and configurable apps. Next, explore distinct printing experiences in ArcGIS Online and ArcGIS Enterprise and crucial troubleshooting tips. Finally, explore techniques for integrating these printing capabilities into your custom web apps built with the ArcGIS Maps SDK for JavaScript.

## SPEAKERS

Noah Sager
Esri

Tanu Hoque
Esri

**Session Type:** Technical Session

**Session Level:** All Attendees

**Access Type:** In-Person, Recorded

**Topic:** Web

**Esri Products:** ArcGIS Enterprise, ArcGIS Online, ArcGIS Pro, ArcGIS Maps SDK for JavaScript

**Capabilities:** Map Output and Printing, Web App Development

**Technologies:** JavaScript, Map Viewer

**Keywords:** print, custom print service, web development, JavaScript



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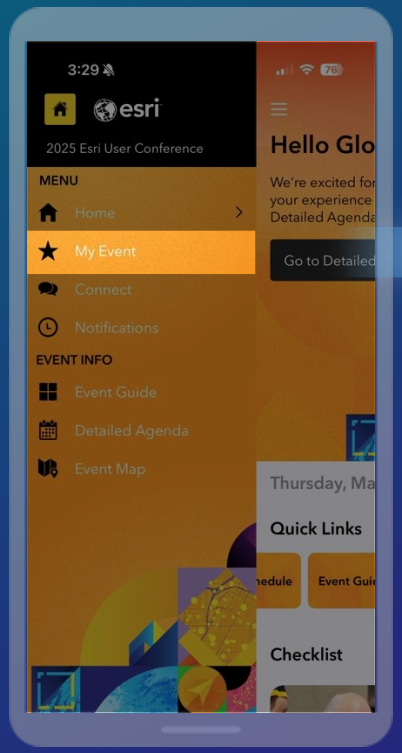
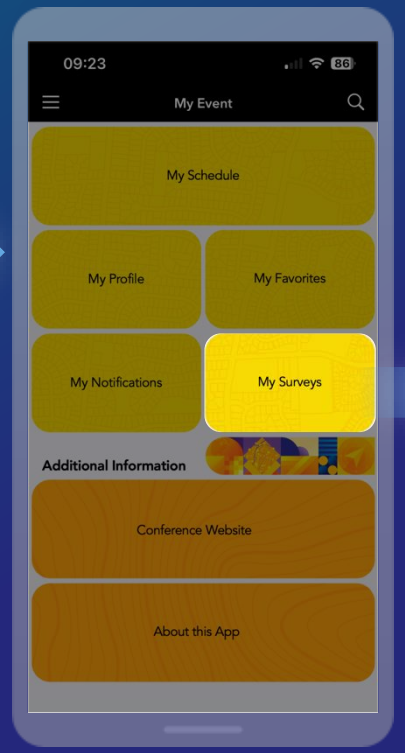
# Please share your feedback in the app

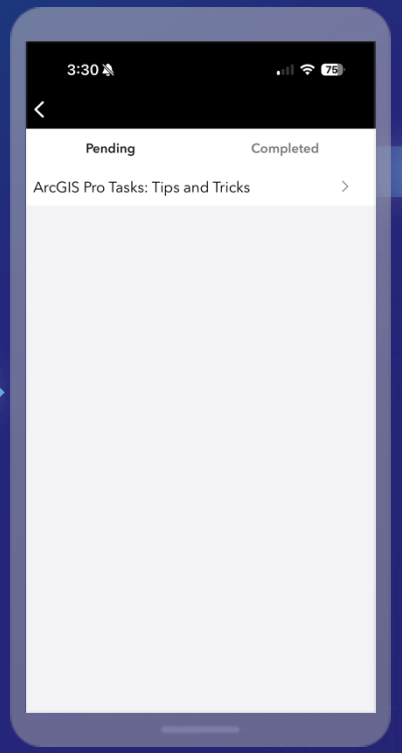
Select “My Event” in the event app main menu



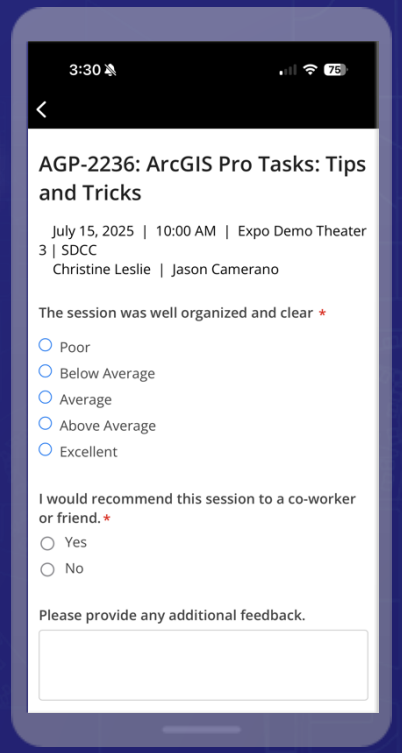

Select “My Surveys”




Select the desired survey




Complete the survey



# Agenda

- Why printing?
- Where does printing come from?
- Does it print?
- Additional resources
- Questions

# Why Printing?

# Why Printing?

- [TODO] – Noah, come-up with several interesting and compelling use cases that help explain why web developers want to do printing…

# Why Printing?

- [TODO] – Noah, come-up with several interesting and compelling use cases that help explain why web developers want to do printing…
- But really, we want to know, why do you print?
- We value your feedback.
- During this session, after this session. Send an email, post on Esri Community.

# Where does Printing come from?

# Where does Printing come from?

ArcGIS Pro

ArcGIS Pro

## Printing begins with ArcGIS Pro

- Print pretty maps
- Make pretty layouts
- Desktop architecture / rules

- Export Webmap GP tool (Print service)
- Share Layout Template
- ArcGIS Online, ArcGIS Enterprise, ArcGIS Server

ArcGIS Pro

# An overview of the Server toolbox

ArcGIS Pro 3.6 | Other versions | Help archive

The Server toolbox contains tools to manage web layers and web maps. It also contains tools that simplify data extraction through the server.

| Toolset | Description |
|---|---|
| Caching | The Caching toolset creates and manages caches for faster display of web tile layers. The caches consist of pregenerated images that the server can distribute to clients. |
| Data Extraction | The Data Extraction toolset provides four tools to implement advanced clip, zip, and ship tasks as an ArcGIS Server geoprocessing service. General extracting tools, such as those in the Extract toolset and the feature analysis tool Clip, only allow you to extract the subset of a single layer data with the output showing the same data format and spatial reference as the input. These tools are somewhat limited for those who want to extract data with the flexibility to select multiple layers, output format, and spatial reference—especially over the Internet. The Data Extraction toolset is designed to facilitate such functionality so you can efficiently extract the multiple-layered data with the specified format and spatial reference through the geoprocessing service. Moreover, you can zip the output file and send it through email, thus making the whole extraction and delivery process more effective. |
| Printing | The Printing toolset contains tools to facilitate printing from web applications. |
| Publishing | The Publishing toolset contains tools to stage, upload, and share web layers and services. |

Toolsets in the Server toolbox

https://pro.arcgis.com/en/pro-app/latest/tool-reference/server/an-overview-of-the-server-toolbox.htm

Publish

ArcGIS Server

ArcGIS Enterprise

ArcGIS Pro

ArcGIS Online

Publish
ArcGIS Server
ArcGIS Enterprise
ArcGIS Pro
ArcGIS Online

ArcGIS Enterprise supports

- Utility print service
- Custom print services
- Custom layouts

ArcGIS Online supports

- Utility print service
- Esri premium print service*
- Custom layouts*

* = depends on user roles and organizational settings

ArcGIS Enterprise supports

- Utility print service
- Custom print services
- Custom layouts

ArcGIS Online supports

- Utility print service
- Esri premium print service*
- Custom layouts*

Publish

* = depends on user roles and organizational settings

ArcGIS P

Note:

Printing from layout items is only available when a template group is configured. Printing from these items will always use the ArcGIS print service, even if you have a custom print service configured. Printing from layout items requires that the services behind the layers in the map are externally available. To print layers secured with web-tier authentication, you must use a custom print service configured to handle web-tier authentication.

Printing from custom layout templates is available to members assigned the Professional or Professional Plus user type.

- Esri premium print service*
- Custom layouts*

ArcGIS Online

ArcGIS

ArcGIS Blog | Mapping | ArcGIS Online | Jun 26, 2024

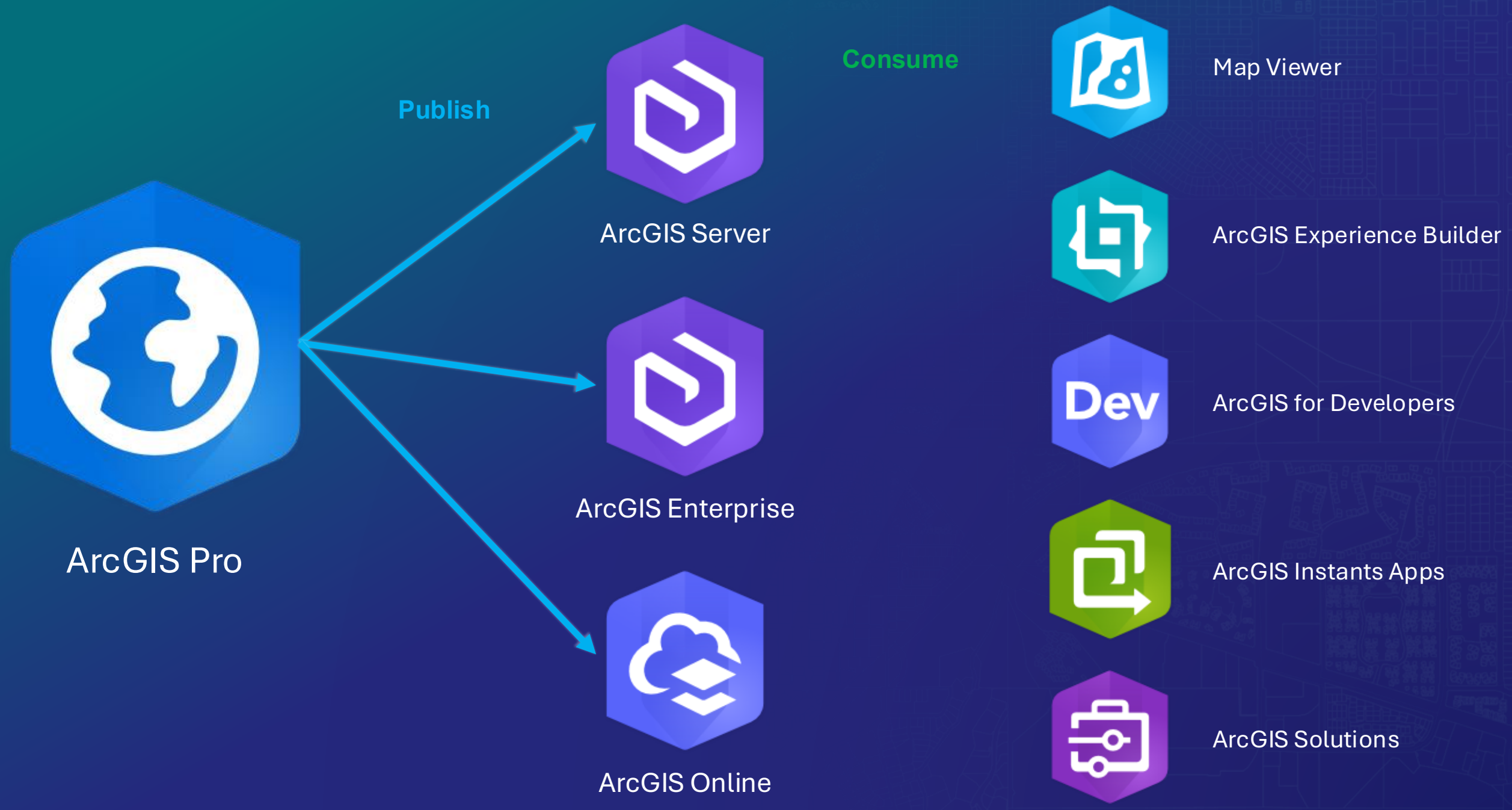
# Customize your print layout experience in ArcGIS Online

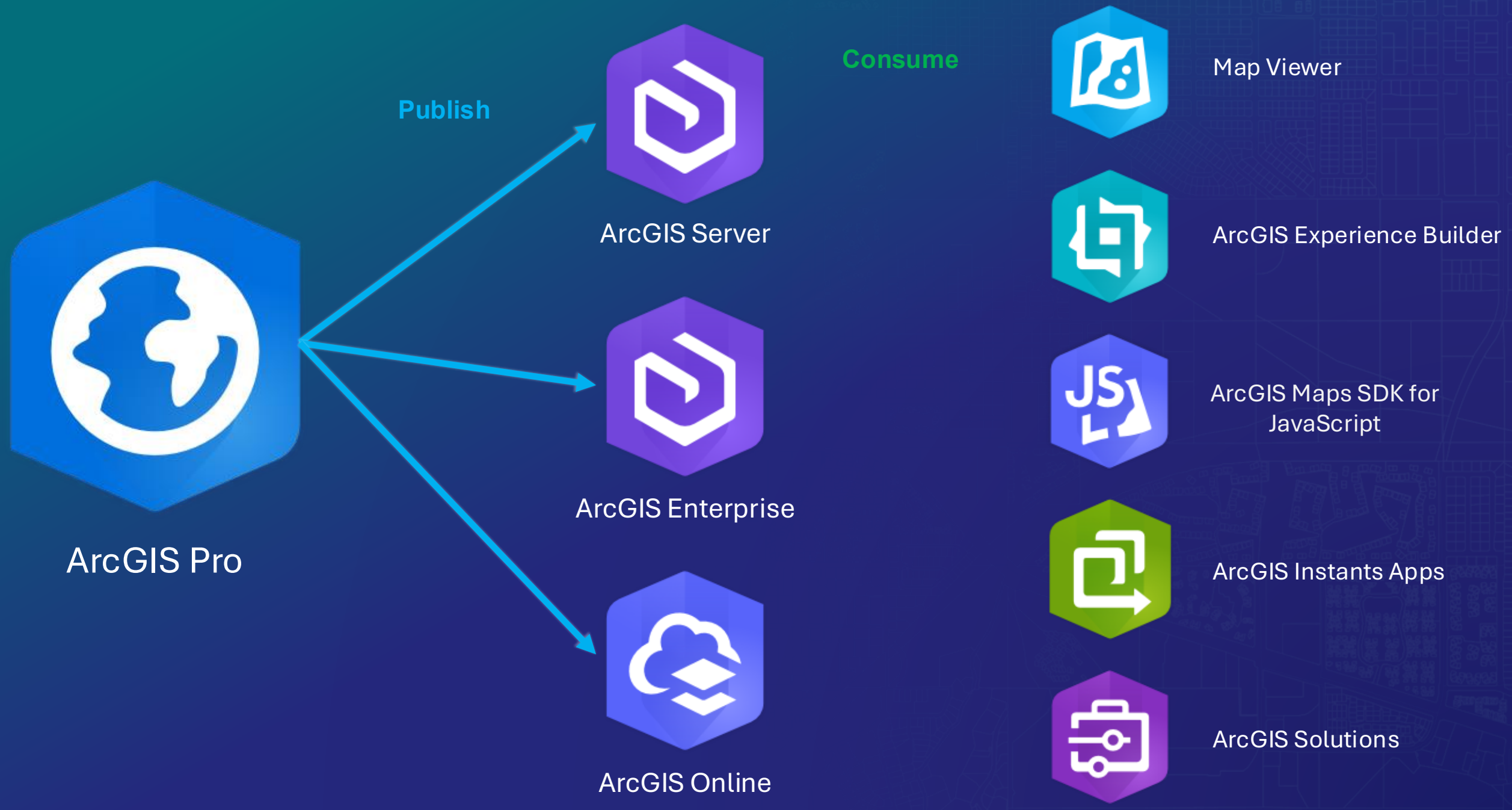
By Zara Matheson

The June 2024 ArcGIS Online update includes the option (beta) to print from layout template items published directly to ArcGIS Online. This enhancement means users can create and publish their own layout items from ArcGIS Pro to ArcGIS Online and use them without having to set up and maintain their own print server. This is especially useful if you want to add your organization's logo or branding to your print output or require more room to support a larger legend. Custom layout items can also add consistency and branding to the print experience for your organization.

This new capability will be available as a beta offering at this time for Professional and Professional Plus user types. Printing from the default templates will continue to be available for all users. Along with this enhancement, the Map Viewer print tool itself has been updated with a cleaner look and we have introduced a print preview option to provide a better indication of the map area that will be included in your print output.

Publish
ArcGIS Server
ArcGIS Enterprise
ArcGIS Pro
ArcGIS Online

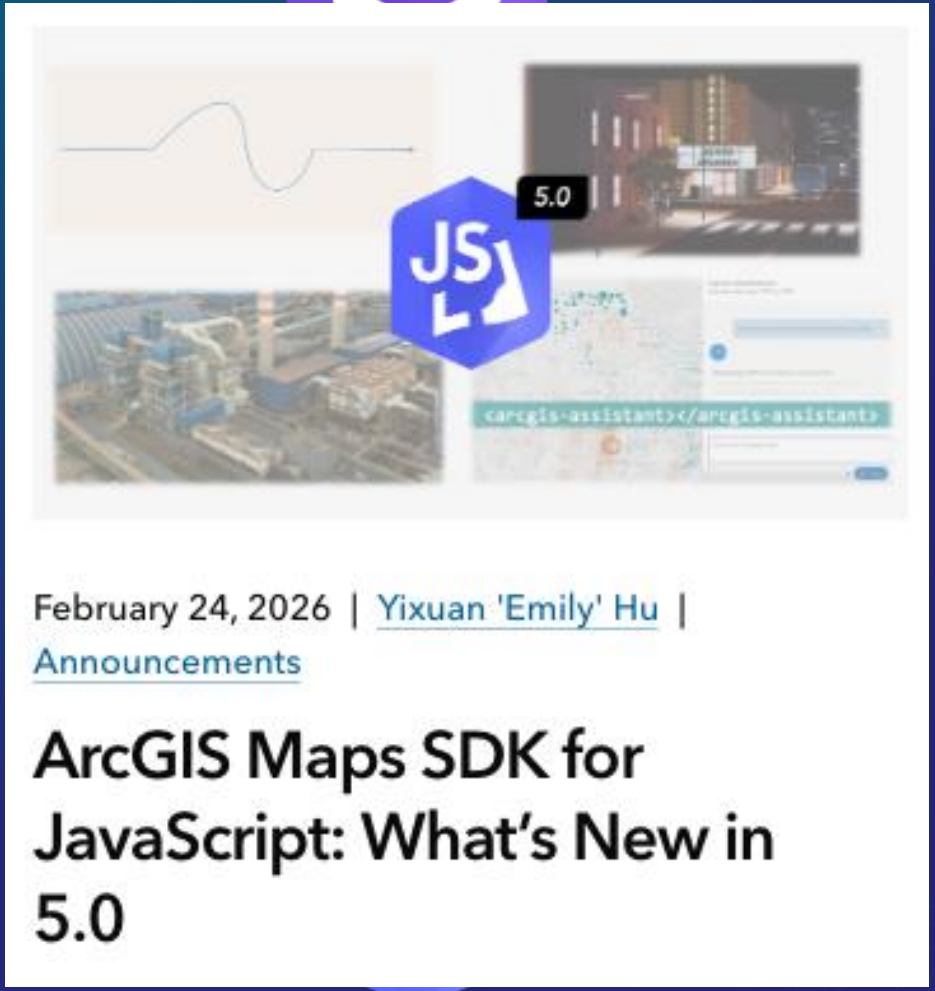

Publish
Consume
ArcGIS Pro
ArcGIS Server
ArcGIS Enterprise
ArcGIS Online
Map Viewer
ArcGIS Experience Builder
Dev
ArcGIS for Developers
ArcGIS Instants Apps
ArcGIS Solutions


Publish
Consume
ArcGIS Pro
ArcGIS Server
ArcGIS Enterprise
ArcGIS Online
Map Viewer
ArcGIS Experience Builder
ArcGIS Maps SDK for JavaScript
ArcGIS Instants Apps
ArcGIS Solutions

Publish

Consume

ArcGIS Pro




ArcGIS Online

Map Viewer

ArcGIS Experience Builder

ArcGIS Maps SDK for JavaScript

ArcGIS Instants Apps

ArcGIS Solutions

ArcGIS Enterprise

ArcGIS Online

**Consume**

Map Viewer

ArcGIS Experience Builder

ArcGIS Maps SDK for JavaScript

ArcGIS Instants Apps

ArcGIS Solutions

**JavaScript Print component**
**Experience Builder Print widget**

ArcGIS Enterprise

**JavaScript Print component**
**Experience Builder Print widget**

ArcGIS Online

Map Viewer

ArcGIS Experience Builder

ArcGIS Maps SDK for JavaScript

ArcGIS Instants Apps

ArcGIS Solutions

**JavaScript Print component**
**Experience Builder Print widget**

ArcGIS Enterprise

**JavaScript Print component**
**Experience Builder Print widget**

ArcGIS Online

**Print component**
Enterprise version
Latest Version

Map Viewer

**Print component**
Enterprise version
Latest Version

ArcGIS Experience Builder

**Print component**
**rest/print ()**
Specific Version

ArcGIS Maps SDK for JavaScript

**Print component**
**rest/print ()**
Specific Version

ArcGIS Instants Apps

**Print component**
**rest/print ()**
Specific Version

ArcGIS Solutions

**JavaScript Print component**
**Experience Builder Print widget**

ArcGIS Enterprise

**JavaScript Print component**
**Experience Builder Print widget**

ArcGIS Online

**Print component**
Enterprise version
Latest Version

Map Viewer

**Print component**
Enterprise version
Latest Version

ArcGIS Experience Builder

**Print component**
**rest/print ()**
Specific Version

ArcGIS Maps SDK for JavaScript

**Print component**
**rest/print ()**
Specific Version

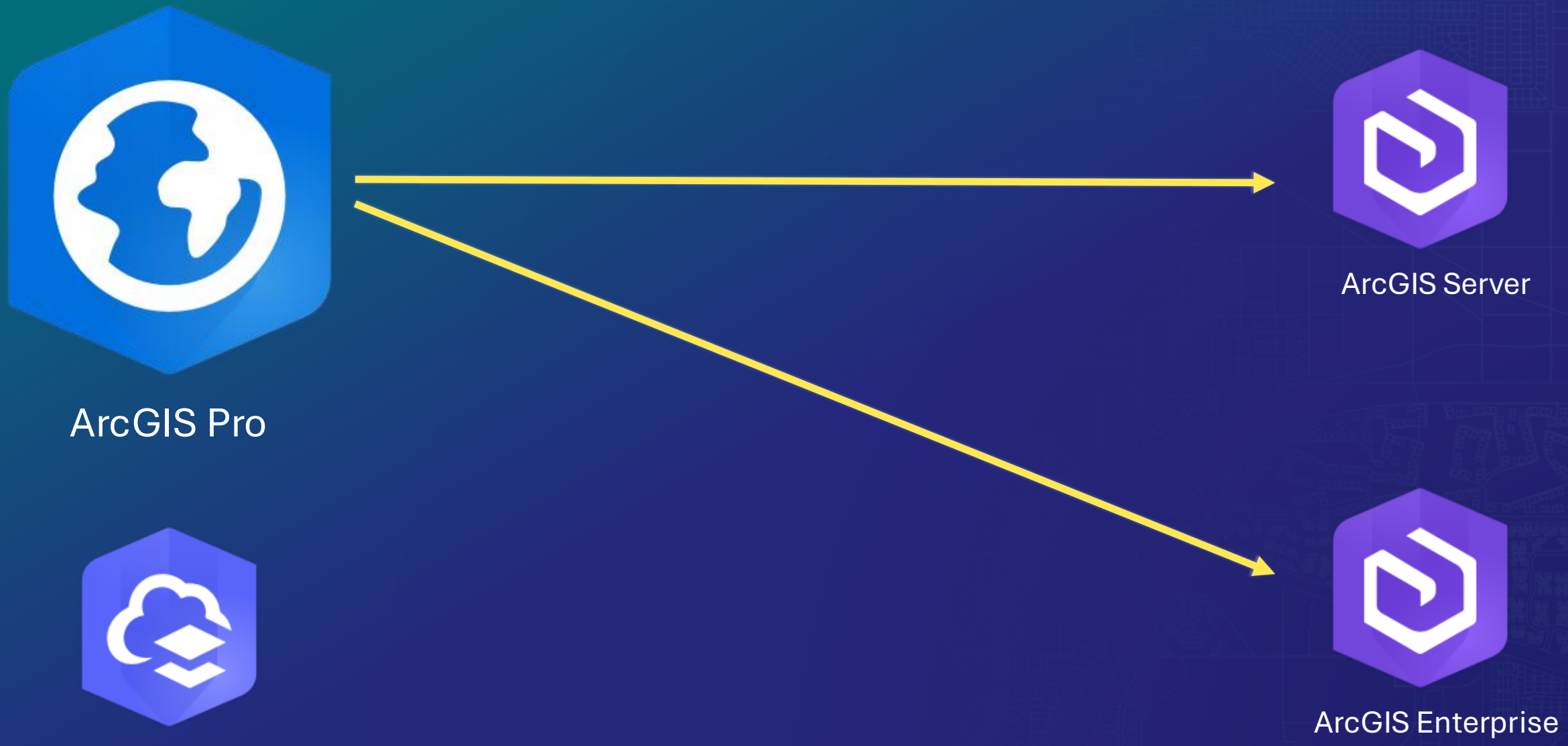
ArcGIS Instants Apps

**Print component**
**rest/print ()**
Specific Version

ArcGIS Solutions

# Publish Custom Services

# Publish Custom Services



# Publish Custom Services

ArcGIS Pro

ArcGIS Server

ArcGIS Enterprise

## Web map printing with arcpy.mp

ArcGIS Pro 3.6 | Other versions | Help archive

A common arcpy.mp workflow used in Portal for ArcGIS is web map printing, in which Python, ArcGIS Experience Builder, ArcGIS Maps SDK for JavaScript and ArcGIS Web AppBuilder work together to create web apps that produce high-quality cartographic output.

# Enable Utility Services

# Enable Utility Services

ArcGIS Server

ArcGIS Enterprise

ArcGIS Online

(No Customization)

Portal for ArcGIS

Home | Use | Administer

Administer / Configure the organization / Utility services

- Fundamentals of configuring the organization
- Best practices for organization maintenance
- Site configuration groups
- Configure general settings
- Configure home page

# Utility services

ArcGIS 11.4 (Windows) | Other versions | Help archive

Utility services enable specific functionality in your organization, for example, printing maps, locating addresses, calculating areas, finding directions, and performing analysis. Some utility services include default services, but you can also u
your organization to use utility services, see Configu

ArcGIS Online

Overview | Features | Resources

Home | Get Started | Create | Analyze | Share | Manage Data | Administer | Reference

Search ArcGIS Online Help

Administer / Configure website

- Manage subscription
- Configure website
  - Configure website
  - Configure general settings

# Configure utility services

Members of the default administrator role and those in custom roles with the appropriate privileges can configure the website to use different utility services to perform printing, geocoding, and routing tasks.

ArcGIS Server

Home | Introduction | Deploy | Manage Data | Manage Services | Administer | Develop

Manage Services / Print services

- Introduction to services
- Publish services
- Work with services
- Edit services
- Migrate services
- Types of services
- Cache services

# Printing in web applications

ArcGIS 11.4 (Windows) | Other versions | Help archive

There are several ways to generate a printable document from a web application. ArcGIS Enterprise includes a geoprocessing service called **PrintingTools**. Web applications invoke the PrintingTools service and get a printable document in return. You can also create a custom print service using your own layouts rather than those in the default PrintingTools service. For advanced printing, you can use a Python script to convert the web map into an ArcGIS Pro Project (.aprx), then export the map or layout inside the project to a variety of formats for printing.

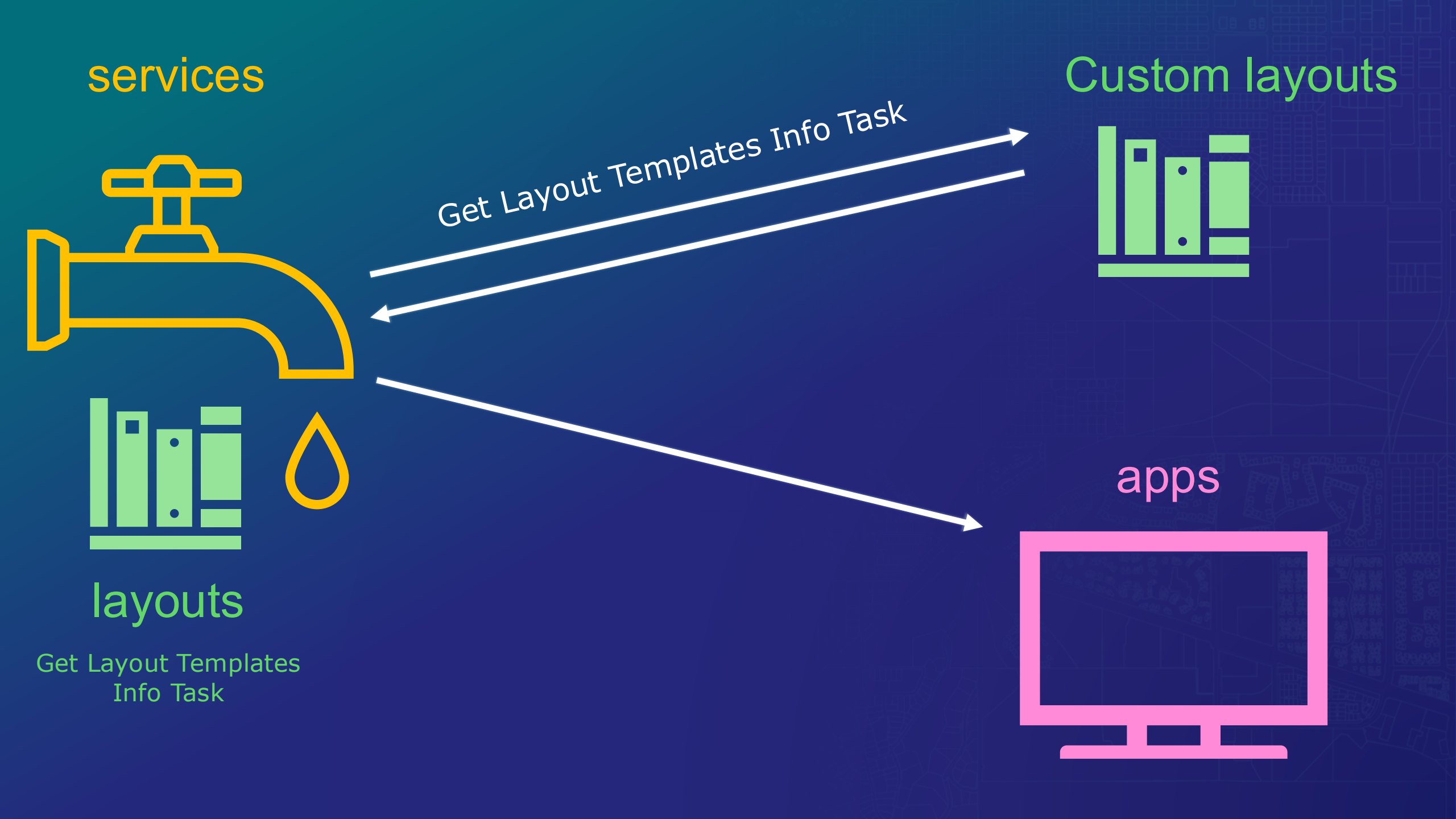
# Share Layout Templates

# Share Layout Templates

The **Get Layout Templates Info** task provides the size of the layout page and the size of a map frame. It also shows whether the following layout elements are available:

- Legend
- Title text
- Copyright text
- Author text
- Custom text elements

You can also add this task to custom print services.


services
Custom layouts
Get Layout Templates Info Task
apps
layouts
Get Layout Templates
Info Task

# Share custom layouts for printing from ArcGIS Pro

ArcGIS 12.0 (Windows) | Other versions | Help archive

When you need printing capabilities in ArcGIS Enterprise outside the scope of the default PrintingTools service, you can create custom layouts in ArcGIS Pro. You can either share these layouts to your portal as a web tool or publish them to a stand-alone ArcGIS Server site as a geoprocessing service. Geoprocessing services that enable printing are called print services.

# Configure custom print layout templates (beta)

Print layout templates contain page setup information for creating printer-friendly documents for maps in Map Viewer. You can publish custom print layout templates as items to ArcGIS Online and share them with a group. You can then configure the group as the organization's layout template group (beta). All custom print layout templates shared with this group are available as template options when printing in Map Viewer.

You can access custom print layout templates that have been published as items to ArcGIS Online and shared with a layout template group (beta) without configuring a print service.

# custom print service

layouts

Get Layout Templates
Info Task

free

Custom
layouts

Get Layout Templates
Info Task

# utility print service

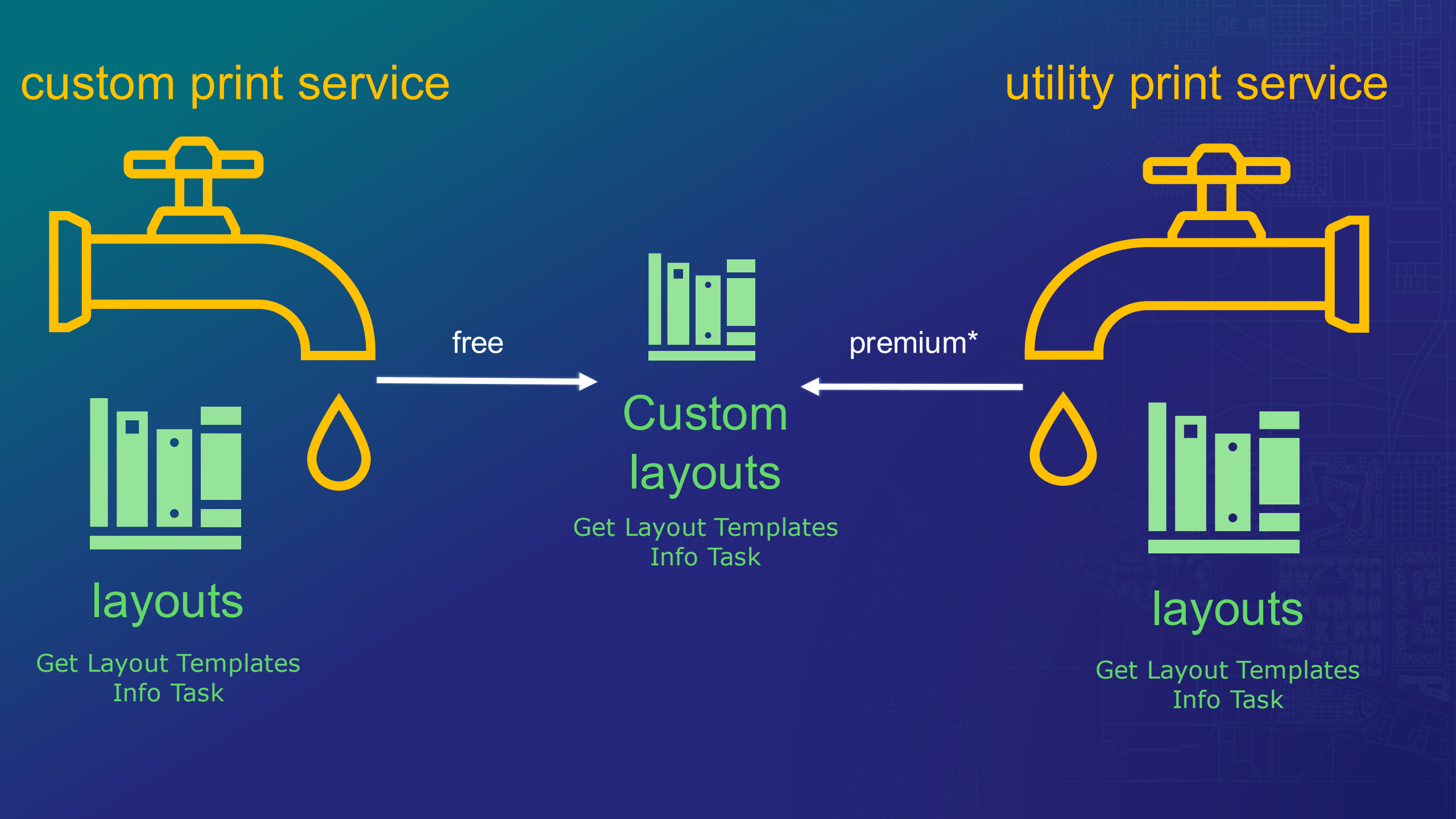
premium*

## Custom layouts

Get Layout Templates
Info Task

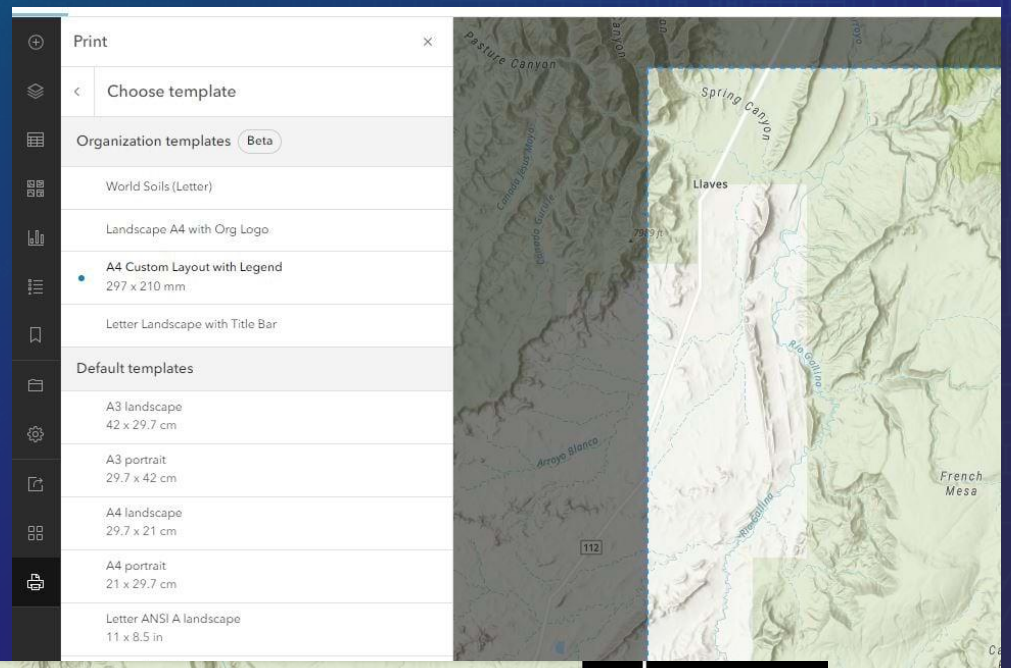
## layouts

Get Layout Templates
Info Task


custom print service
utility print service
free
premium*
Custom layouts
Get Layout Templates Info Task
layouts
Get Layout Templates Info Task
layouts
Get Layout Templates Info Task

# Customize your print layout experience in ArcGIS Online

By Zara Matheson



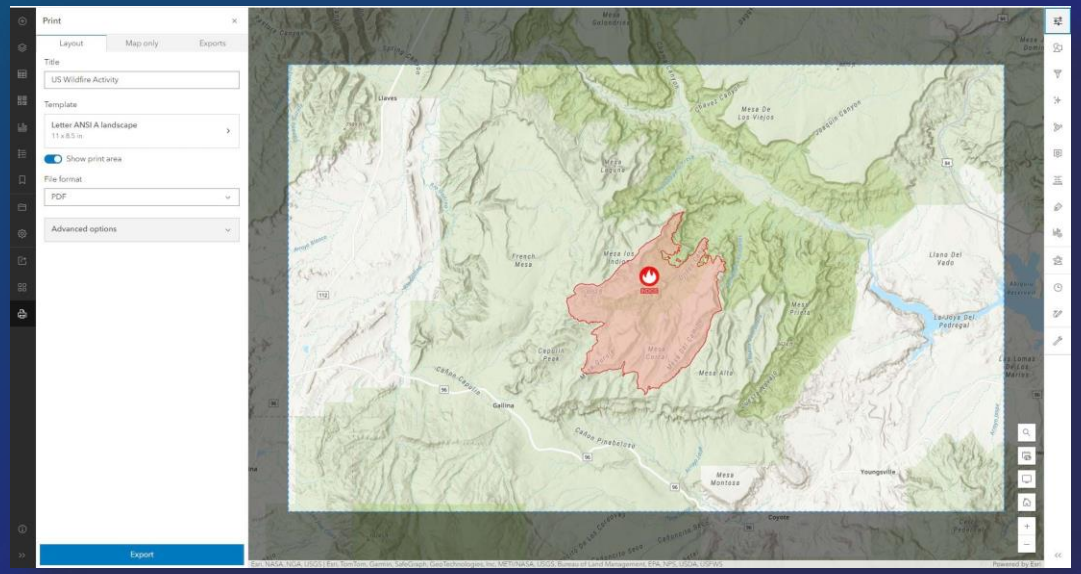

Layout template group

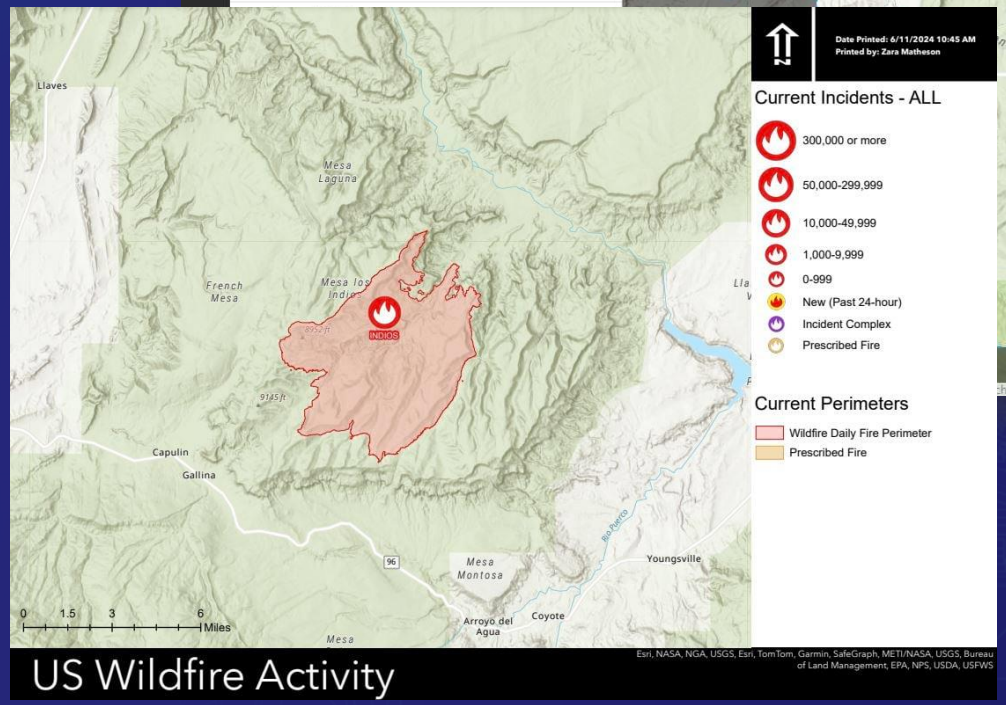
Layout items contain page setup information for creating printer-friendly documents of your maps. Web apps like the Map Viewer can use these layouts as templates for printing web maps. Choose which group contains the layouts you want to use for printing in your organization.

M My Layout Template Group

Sort order

Title | Ascending




US Wildfire Activity

# How?

ArcGIS Pro

Tanu Hoque

# Does it print?

Map Viewer

ArcGIS Experience Builder

ArcGIS Maps SDK for JavaScript

Noah Sager

# Does it print?

Map Viewer

ArcGIS Experience Builder

ArcGIS Maps SDK for JavaScript

Noah Sager

# Demos

# Printing for Web Developers

## Additional Resources

***ESRI DEVELOPER & TECHNOLOGY SUMMIT***

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# ArcGIS Enterprise: Printing Reports and Maps from ArcGIS Web Applications

**LIVE IN 1 HOUR**

Friday, Mar 13 | 10:00 AM - 11:00 AM PDT (12:00 PM - 1:00 PM CDT) | Primrose C | PSCC

SCHEDULED FAVORITED

SCHEDULED Friday, Mar 13 | 10:00 AM - 11:00 AM PDT (12:00 PM - 1:00 PM CDT) | Primrose C | PSCC | Available Online

Learn how to generate printable reports and maps from the ArcGIS Enterprise print service, which was enhanced at version 11.2. Reports can be generated from templates saved as portal items or referenced by the print service, and can include maps, attributes, attachments, related records and more. In this technical session, explore how you can configure layout and report templates in ArcGIS Pro, share your own print service with reporting capabilities, and use the service in Map Viewer, ArcGIS Experience Builder, or custom apps built with ArcGIS Maps SDK for JavaScript.

## SPEAKERS

Tanu Hoque
Esri

Jill Scholz
Esri

**Session Type:** Technical Session

**Session Level:** Intermediate

**Access Type:** In-Person, Recorded, Live stream

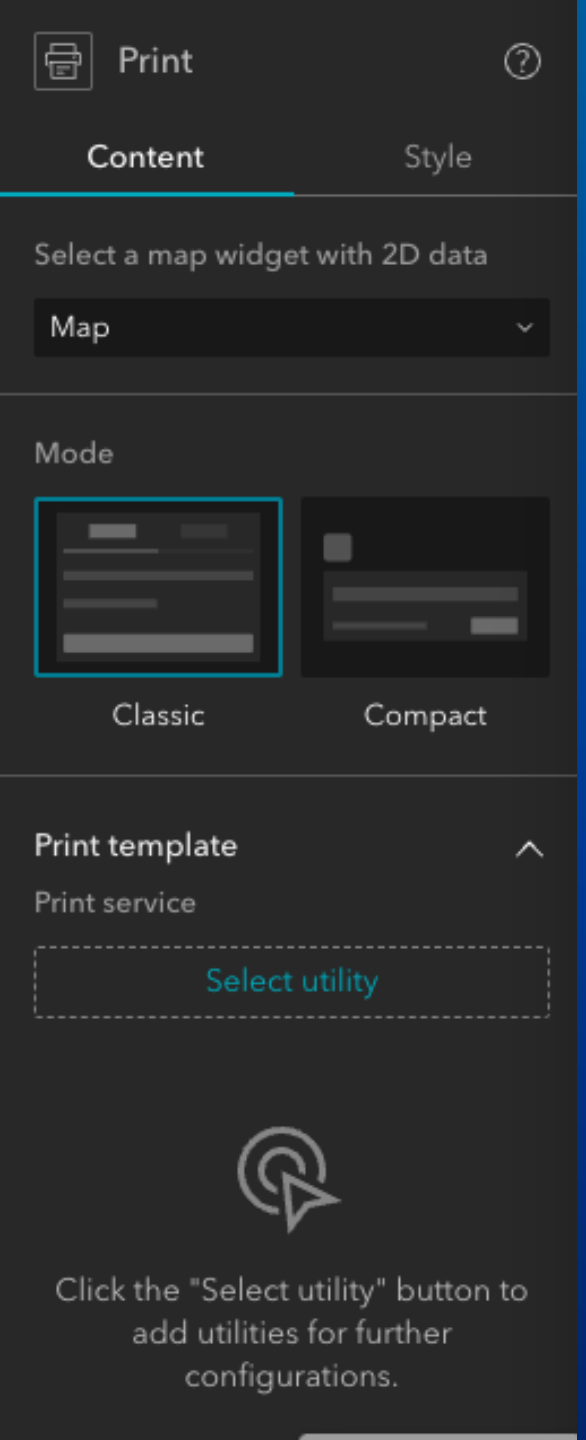
**Topic:** ArcGIS Enterprise

**Esri Products:** ArcGIS Experience Builder, ArcGIS Pro, Map Viewer

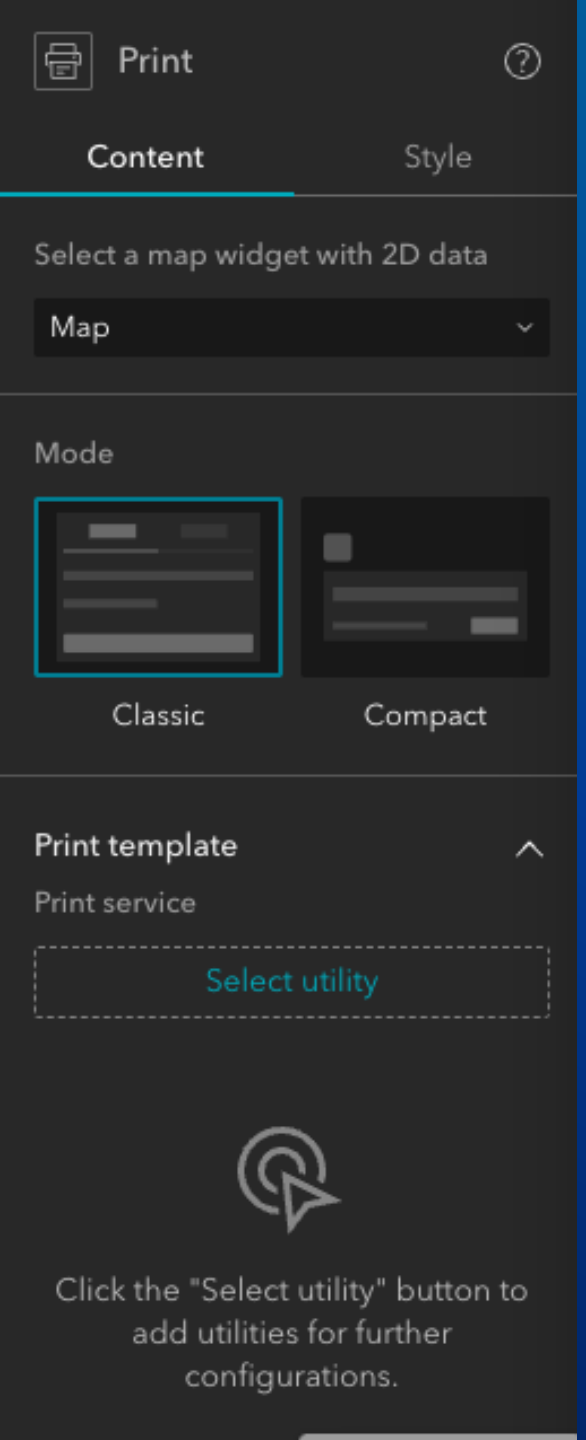
**Capabilities:** Map Output and Printing

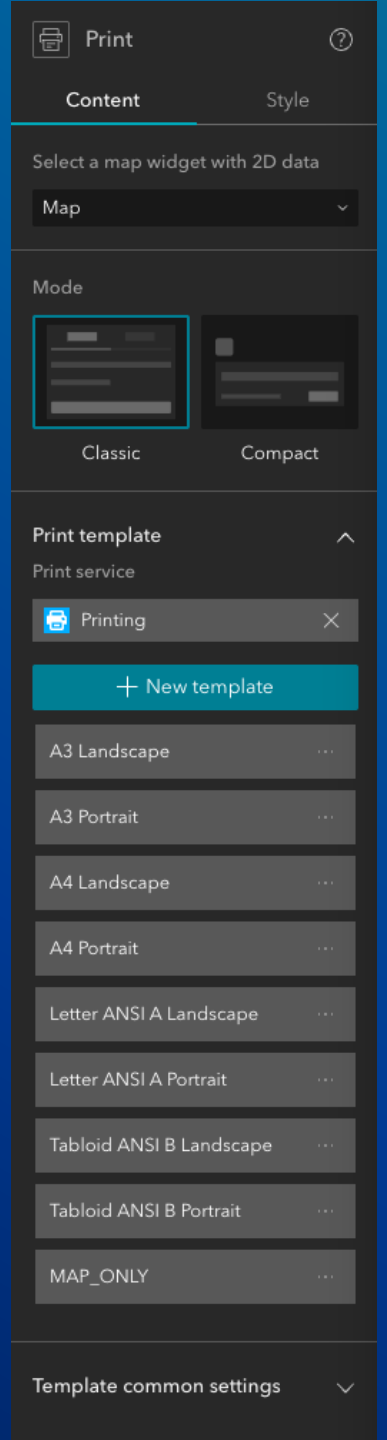
**Technologies:** Cross-Platform

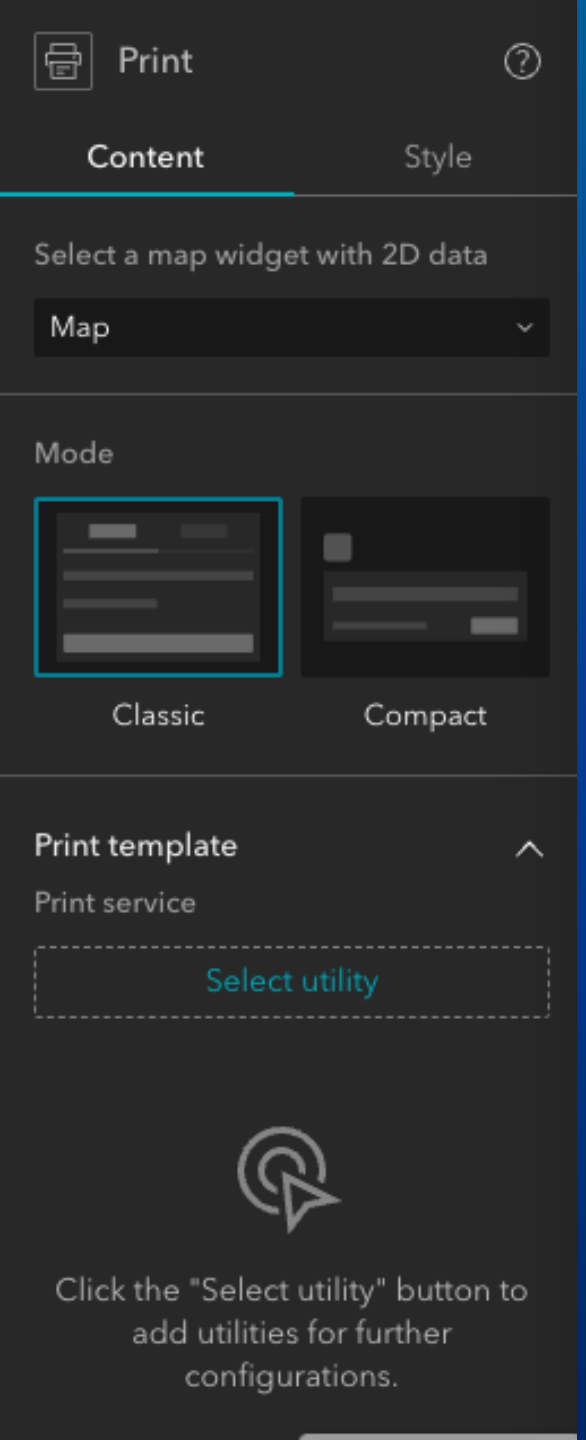
**Keywords:** reports, printing, configure, templates

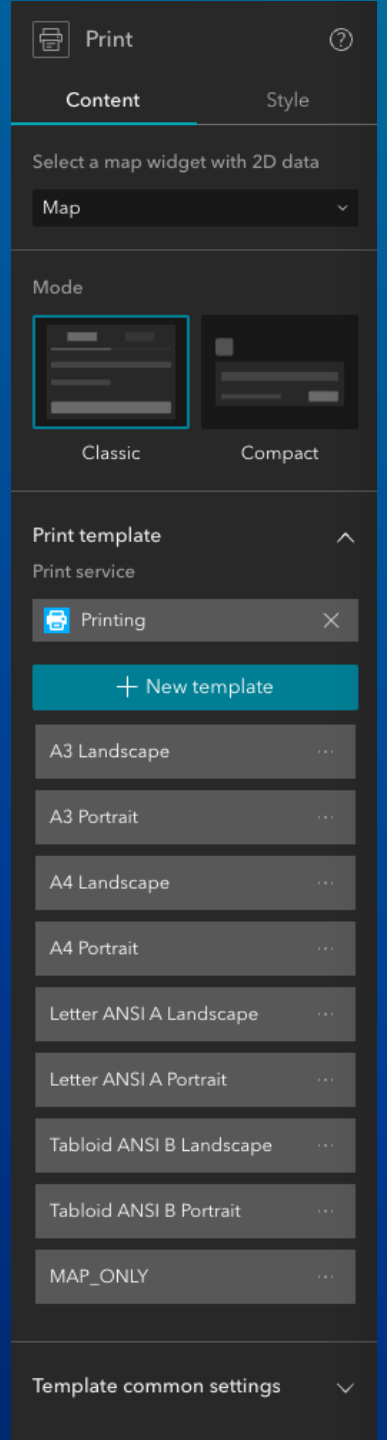



# Experience Builder


Print
Content
Style
Select a map widget with 2D data
Map
Mode
Classic
Compact
Print template
Print service
Select utility
Click the "Select utility" button to add utilities for further configurations.


Print
Content
Style
Select a map widget with 2D data
Map
Mode
Classic
Compact
Print template
Print service
Printing
New template
A3 Landscape
A3 Portrait
A4 Landscape
A4 Portrait
Letter ANSI A Landscape
Letter ANSI A Portrait
Tabloid ANSI B Landscape
Tabloid ANSI B Portrait
MAP_ONLY
Template common settings


Print
Content
Style
Select a map widget with 2D data
Map
Mode
Classic
Compact
Print template
Print service
Select utility
Click the "Select utility" button to add utilities for further configurations.


Print
Content
Style
Select a map widget with 2D data
Map
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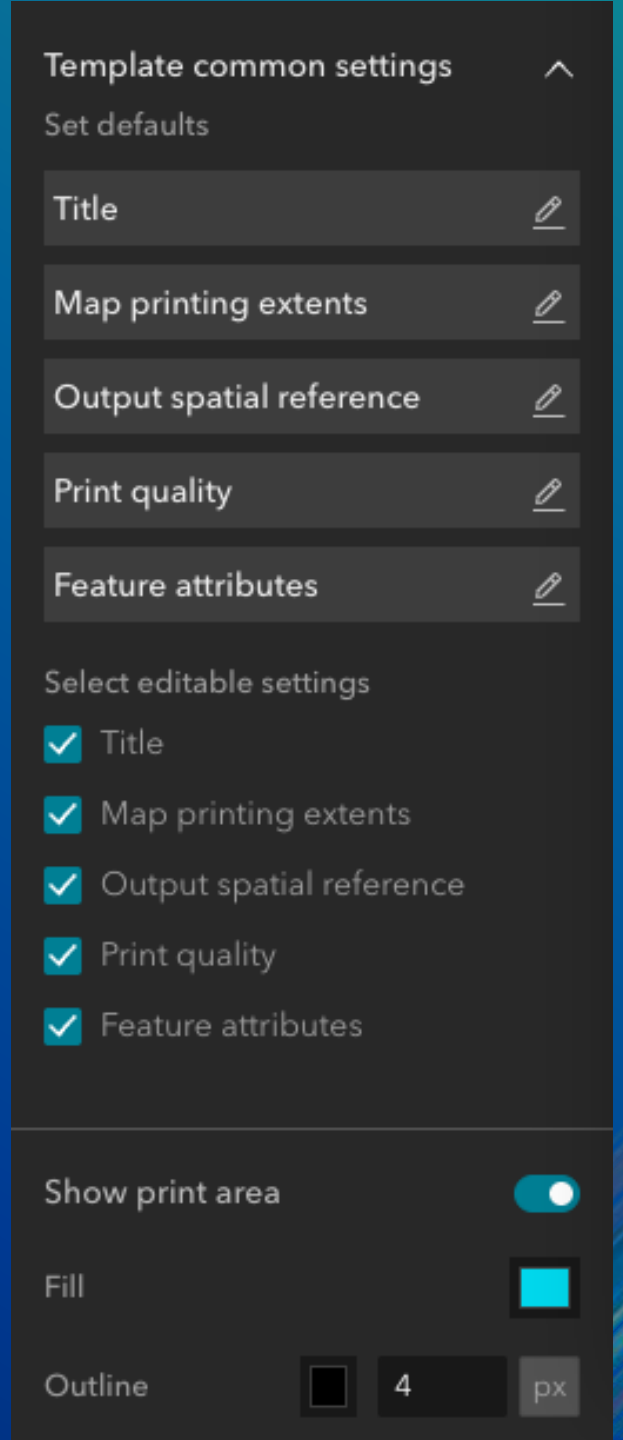
Compact
Print template
Print service
Printing
New template
A3 Landscape
A3 Portrait
A4 Landscape
A4 Portrait
Letter ANSI A Landscape
Letter ANSI A Portrait
Tabloid ANSI B Landscape
Tabloid ANSI B Portrait
MAP_ONLY
Template common settings


Template common settings
Set defaults
Title
Map printing extents
Output spatial reference
Print quality
Feature attributes
Select editable settings
Title
Map printing extents
Output spatial reference
Print quality
Feature attributes
Show print area
Fill
Outline
4
px

Print

Content | Style

Select a map widget with 2D data

Map

Mode

Classic | Compact

Print template

Print service

Select utility

Click the "Select utility" button to add utilities for further configurations.

Print

Content | Style

Select a map widget with 2D data

Map

Mode

Classic | Compact

Print template

Print service

Printing

+ New template

- A3 Landscape
- A3 Portrait
- A4 Landscape
- A4 Portrait
- Letter ANSI A Landscape
- Letter ANSI A Portrait
- Tabloid ANSI B Landscape
- Tabloid ANSI B Portrait
- MAP_ONLY

Template common settings

Template common settings

Set defaults

- Title
- Map printing extents
- Output spatial reference
- Print quality
- Feature attributes

Select editable settings

- [x] Title
- [x] Map printing extents
- [x] Output spatial reference
- [x] Print quality
- [x] Feature attributes

Show print area

Fill

Outline 4 px

Template configuration

Template name

Printing for Web Developers

File format

pdf

Layout

A3 Landscape

Layout options

Set defaults

- Author
- Copyright
- Legend
- Scalebar unit
- North Arrow

Select editable settings

- [x] Author
- [x] Copyright
- [x] Legend
- [x] Scalebar unit
- [x] North arrow

Override common settings

# Community

## Dedicated support

### Esri Technical Support

Esri Technical Support is available using live chat, phone or email (with a paid subscription plan).

## Additional support

### Videos

Learn from the pros and the community.

### Esri Community

Ask questions, share ideas, and collaborate with others on the ArcGIS Maps SDK for JavaScript.

### Certification exam

Validate your skills and expertise on the ArcGIS Maps SDK for JavaScript. Demonstrate proficiency using the SDK for mapping, visualization, handling asynchronous events, and integrating with components.

### Esri Academy

Check out the Getting Started with ArcGIS Maps SDK for JavaScript class on Esri Academy, designed for beginners in GIS and web development.

### Stack Overflow

Join a large and growing community of developers to find the best answers to coding questions. When asking a question, remember to include the "ArcGIS" tag for others to find your questions about the ArcGIS Maps SDK for JavaScript.

### GIS Stack Exchange

A huge community of GIS professionals helping each other with Geographic Information System software questions.

https://developers.arcgis.com/javascript/latest/community/

# Troubleshooting

## Five triage steps

For issues that only occur when building applications locally on your computer, there are a few steps you can take to help isolate the cause:

**Step 1 - Check for error messages.** Look for any error messages or warnings in the terminal window when building the application, and in the browser's console when running the application. These messages provide important clues about where the problem is occurring, such as whether or not it is originating from inside an SDK module. Also, take note of when the error happens, such as during the build process or when the application is running, and you click on the map.

**Step 2 - Search for existing issues.** Before reporting a new issue, it's a good idea to search for existing issues on the Esri Community site or on the internet. Someone else may have already encountered and resolved a similar problem. Also, consider reviewing the SDK release notes for breaking changes and bug fixes and enhancements.

**Step 3 - Check the documentation.** Make sure you are using the SDK's API reference to verify that you are using the API correctly. You may have missed a step or misconfigured something, such as a property or method.

**Step 4 - Verify the project configuration.** Compare your project configuration to the sample applications on the jsapi-resources GitHub repository. The samples can offer hints and suggestions on best practices. For example, there are examples of importing CSS and setting up TypeScript.

**Step 5 - Update dependencies.** Where possible, ensure that you have the latest version of the SDK and any other npm package dependencies being used in the project. Outdated dependencies can sometimes cause compatibility issues. Also, upgrading to the latest version of the SDK can frequently resolve functionality issues.

# Complex vs. Complicated

# Complex vs. Complicated

A cat is complex:
Interconnected, interdependent parts that are unpredictable when put together

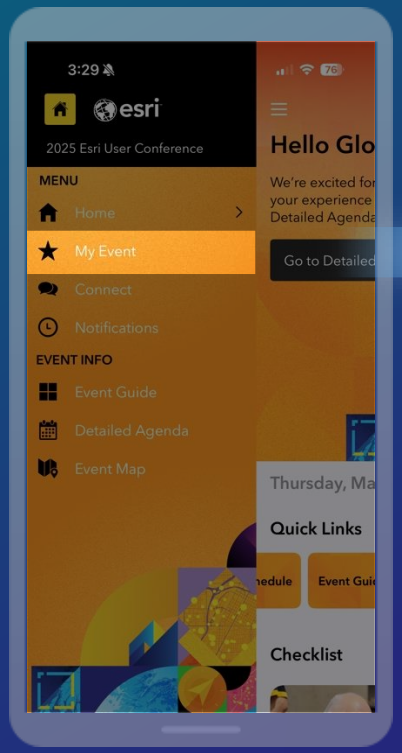
# Complex vs. Complicated

A cat is complex:
Interconnected, interdependent parts that are unpredictable when put together

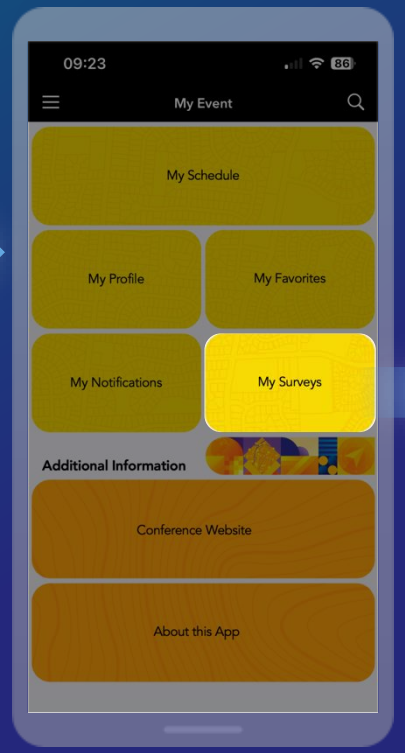
Printing is complicated:
Intricate, multi-layered system that is rewarding to understand and generally predictable

# Please share your feedback in the app

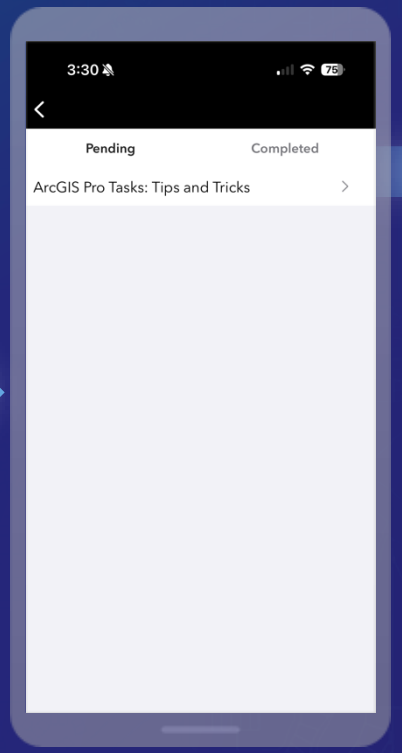
Select “My Event” in the event app main menu



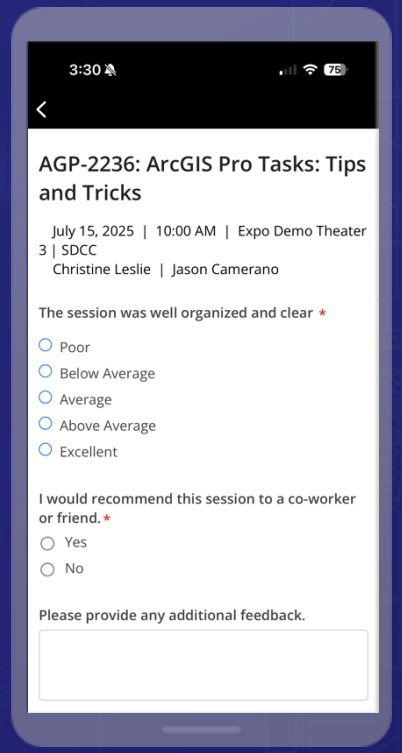

Select “My Surveys”




Select the desired survey




Complete the survey



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# Questions

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